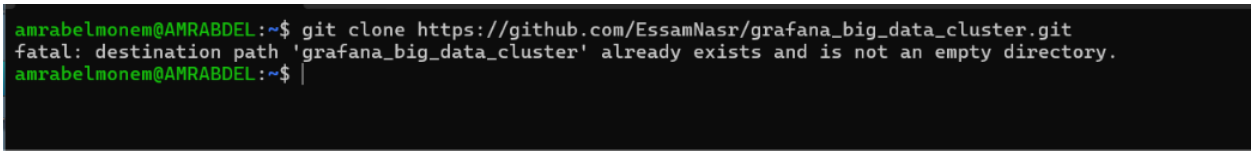
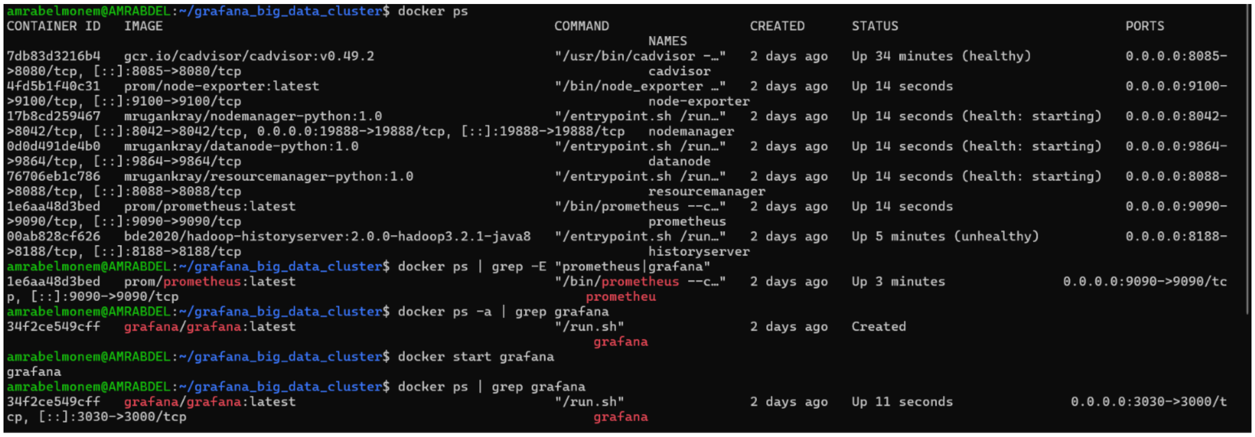
1-open docker and open wsl and clone eng essam repo



```
amrabelmonem@AMRABDEL:~$ git clone https://github.com/EssamNasr/grafana_big_data_cluster.git
fatal: destination path 'grafana_big_data_cluster' already exists and is not an empty directory.
amrabelmonem@AMRABDEL:~$ |
```

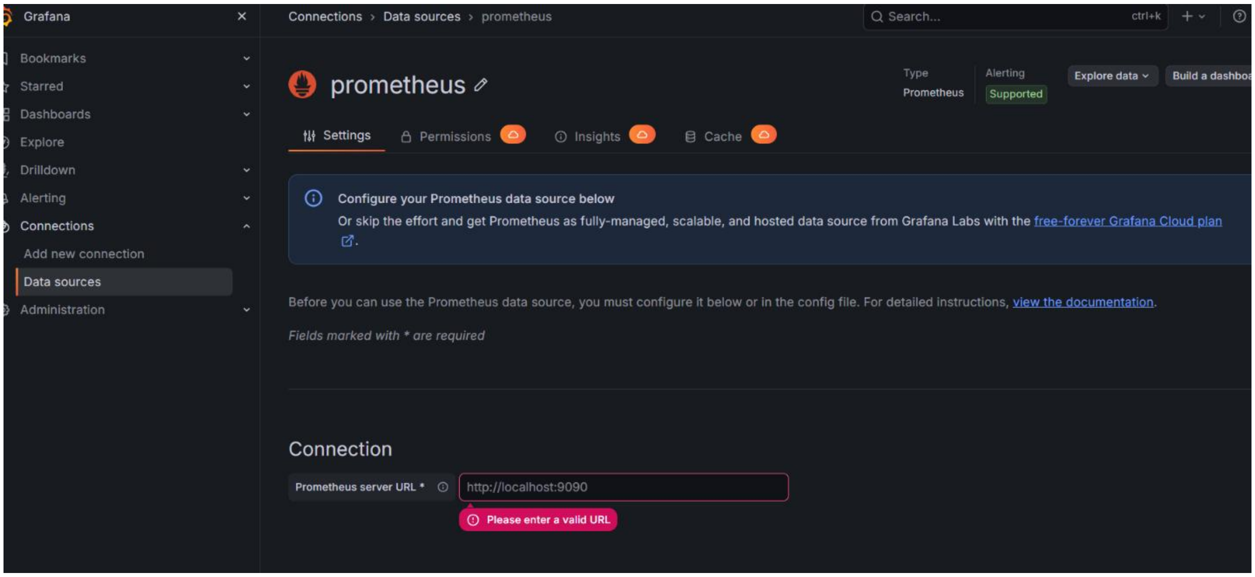
Fix Prometheus and Grafana



```
amrabelmonem@AMRABDEL:~/grafana_big_data_cluster$ docker ps
CONTAINER ID   IMAGE                                                  COMMAND                  CREATED      STATUS                             PORTS
                                                                                  NAMES
7db83d3216b4   gcr.io/cadvisor/cadvisor:v0.49.2                       "/usr/bin/cadvisor -…"   2 days ago   Up 34 minutes (healthy)            0.0.0.0:8085-
>8080/tcp, [::]:8085->8080/tcp                                                    cadvisor
4fd5b1f40c31   prom/node-exporter:latest                              "/bin/node_exporter …"   2 days ago   Up 14 seconds                      0.0.0.0:9100-
>9100/tcp, [::]:9100->9100/tcp                                                    node-exporter
17b8cd259467   mrugankray/nodemanager-python:1.0                      "/entrypoint.sh /run…"   2 days ago   Up 14 seconds (health: starting)   0.0.0.0:8042-
>8042/tcp, [::]:8042->8042/tcp, 0.0.0.0:19888->19888/tcp, [::]:19888->19888/tcp   nodemanager
0d0d491de4b0   mrugankray/datanode-python:1.0                         "/entrypoint.sh /run…"   2 days ago   Up 14 seconds (health: starting)   0.0.0.0:9864-
>9864/tcp, [::]:9864->9864/tcp                                                    datanode
76706eb1c786   mrugankray/resourcemanager-python:1.0                  "/entrypoint.sh /run…"   2 days ago   Up 14 seconds (health: starting)   0.0.0.0:8088-
>8088/tcp, [::]:8088->8088/tcp                                                    resourcemanager
1e6aa48d3bed   prom/prometheus:latest                                 "/bin/prometheus --c…"   2 days ago   Up 14 seconds                      0.0.0.0:9090-
>9090/tcp, [::]:9090->9090/tcp                                                    prometheus
00ab828cf626   bde2020/hadoop-historyserver:2.0.0-hadoop3.2.1-java8   "/entrypoint.sh /run…"   2 days ago   Up 5 minutes (unhealthy)           0.0.0.0:8188-
>8188/tcp, [::]:8188->8188/tcp                                                    historyserver
amrabelmonem@AMRABDEL:~/grafana_big_data_cluster$ docker ps | grep -E "prometheus|grafana"
1e6aa48d3bed   prom/prometheus:latest                                 "/bin/prometheus --c…"   2 days ago   Up 3 minutes               0.0.0.0:9090->9090/tc
p, [::]:9090->9090/tcp                                                    prometheu
amrabelmonem@AMRABDEL:~/grafana_big_data_cluster$ docker ps -a | grep grafana
34f2ce549cff   grafana/grafana:latest                                 "/run.sh"                2 days ago   Created
                                                                           grafana

amrabelmonem@AMRABDEL:~/grafana_big_data_cluster$ docker start grafana
grafana
amrabelmonem@AMRABDEL:~/grafana_big_data_cluster$ docker ps | grep grafana
34f2ce549cff   grafana/grafana:latest                                 "/run.sh"                2 days ago   Up 11 seconds               0.0.0.0:3030->3000/t
cp, [::]:3030->3000/tcp                                                    grafana
```

Add promethus in Grafana



Configure your Prometheus data source below

Or skip the effort and get Prometheus as fully-managed, scalable, and hosted data source from Grafana Labs with the free-forever Grafana Cloud plan.

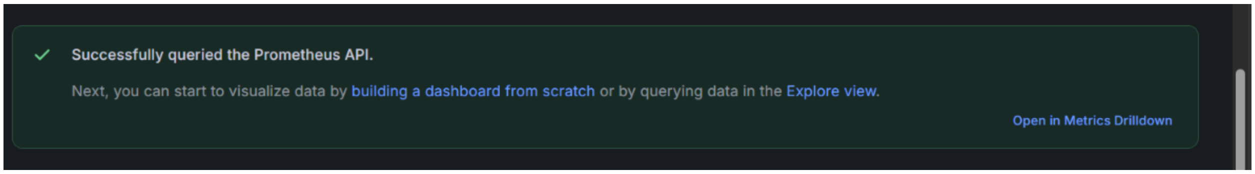
Before you can use the Prometheus data source, you must configure it below or in the config file. For detailed instructions, view the documentation.

*Fields marked with * are required*

Connection

Prometheus server URL * http://localhost:9090

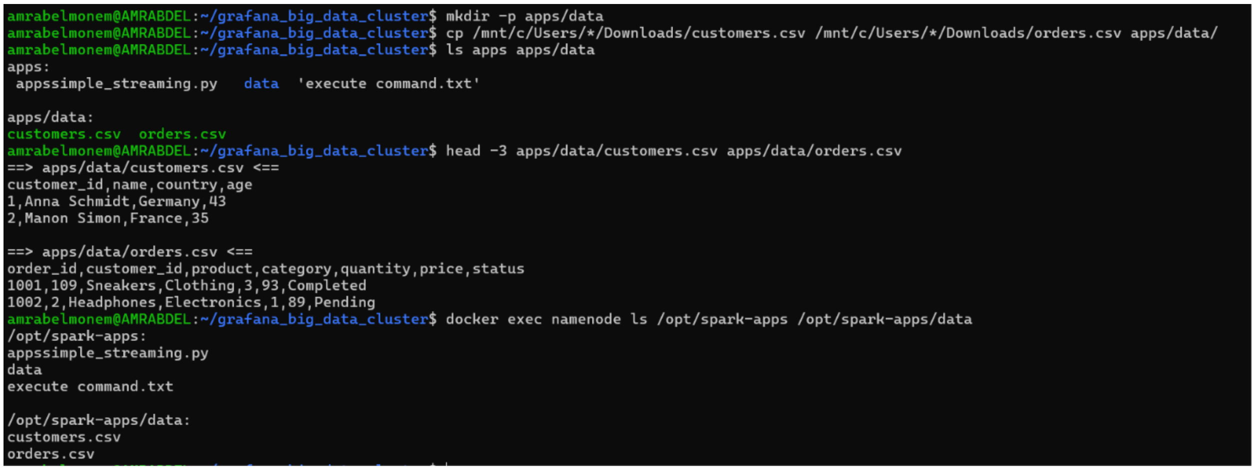
Please enter a valid URL



Successfully queried the Prometheus API.

Next, you can start to visualize data by building a dashboard from scratch or by querying data in the Explore view.

make data folder and spark application



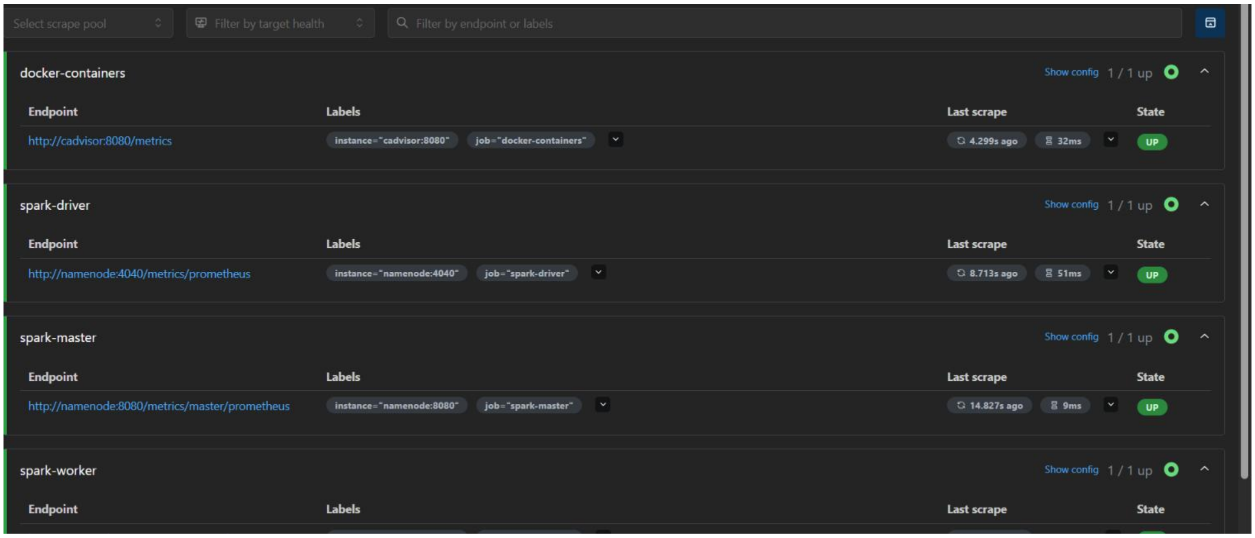
```
amrabelmonem@AMRABDEL:~/grafana_big_data_cluster$ mkdir -p apps/data
amrabelmonem@AMRABDEL:~/grafana_big_data_cluster$ cp /mnt/c/Users/*/Downloads/customers.csv /mnt/c/Users/*/Downloads/orders.csv apps/data/
amrabelmonem@AMRABDEL:~/grafana_big_data_cluster$ ls apps apps/data
apps:
 appssimple_streaming.py   data  'execute command.txt'

apps/data:
customers.csv  orders.csv
amrabelmonem@AMRABDEL:~/grafana_big_data_cluster$ head -3 apps/data/customers.csv apps/data/orders.csv
==> apps/data/customers.csv <==
customer_id,name,country,age
1,Anna Schmidt,Germany,43
2,Manon Simon,France,35

==> apps/data/orders.csv <==
order_id,customer_id,product,category,quantity,price,status
1001,109,Sneakers,Clothing,3,93,Completed
1002,2,Headphones,Electronics,1,89,Pending
amrabelmonem@AMRABDEL:~/grafana_big_data_cluster$ docker exec namenode ls /opt/spark-apps /opt/spark-apps/data
/opt/spark-apps:
appssimple_streaming.py
data
execute command.txt

/opt/spark-apps/data:
customers.csv
orders.csv
```

Prometheus Targets



docker-containers

Show config 1 / 1 up

Endpoint	Labels	Last scrape	State
http://cadvisor:8080/metrics	instance="cadvisor:8080" job="docker-containers"	4.299s ago 32ms	UP

spark-driver

Show config 1 / 1 up

Endpoint	Labels	Last scrape	State
http://namenode:4040/metrics/prometheus	instance="namenode:4040" job="spark-driver"	8.713s ago 51ms	UP

spark-master

Show config 1 / 1 up

Endpoint	Labels	Last scrape	State
http://namenode:8080/metrics/master/prometheus	instance="namenode:8080" job="spark-master"	14.827s ago 9ms	UP

spark-worker

Show config 1 / 1 up

Endpoint	Labels	Last scrape	State

Spark Cluster Monitoring